

Digital Transmission of Analog Signals

Digitization

Pulse Code Modulation (PCM)

Complied By:

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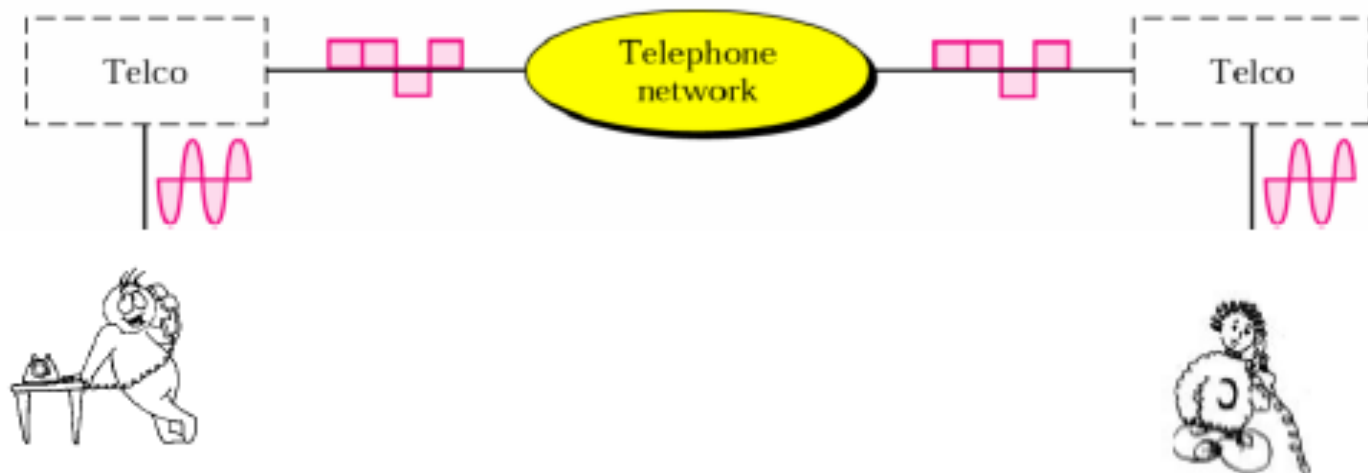
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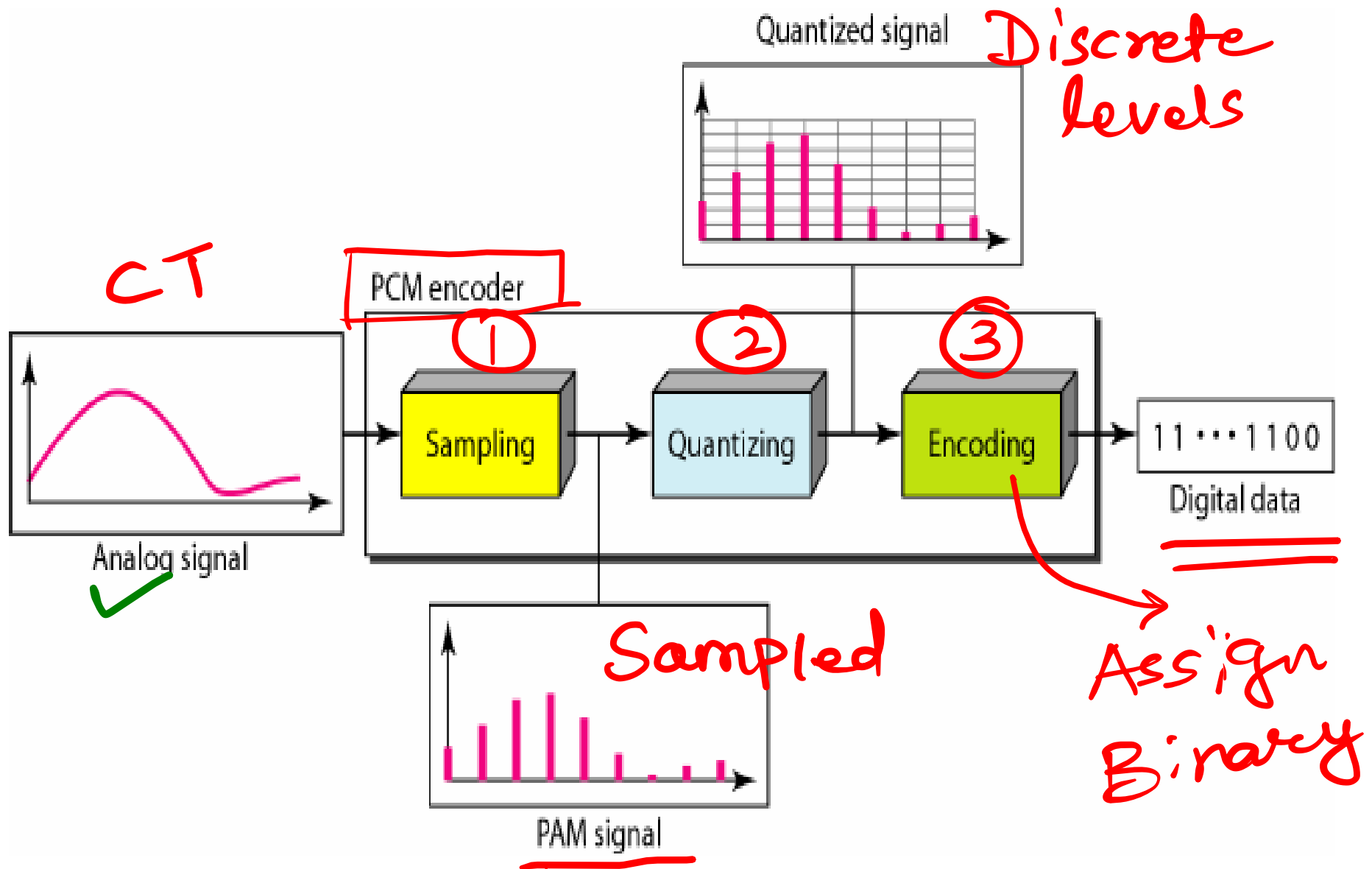
Digital Transmission of Analog signal

Digitization – process of converting analog data into digital signal

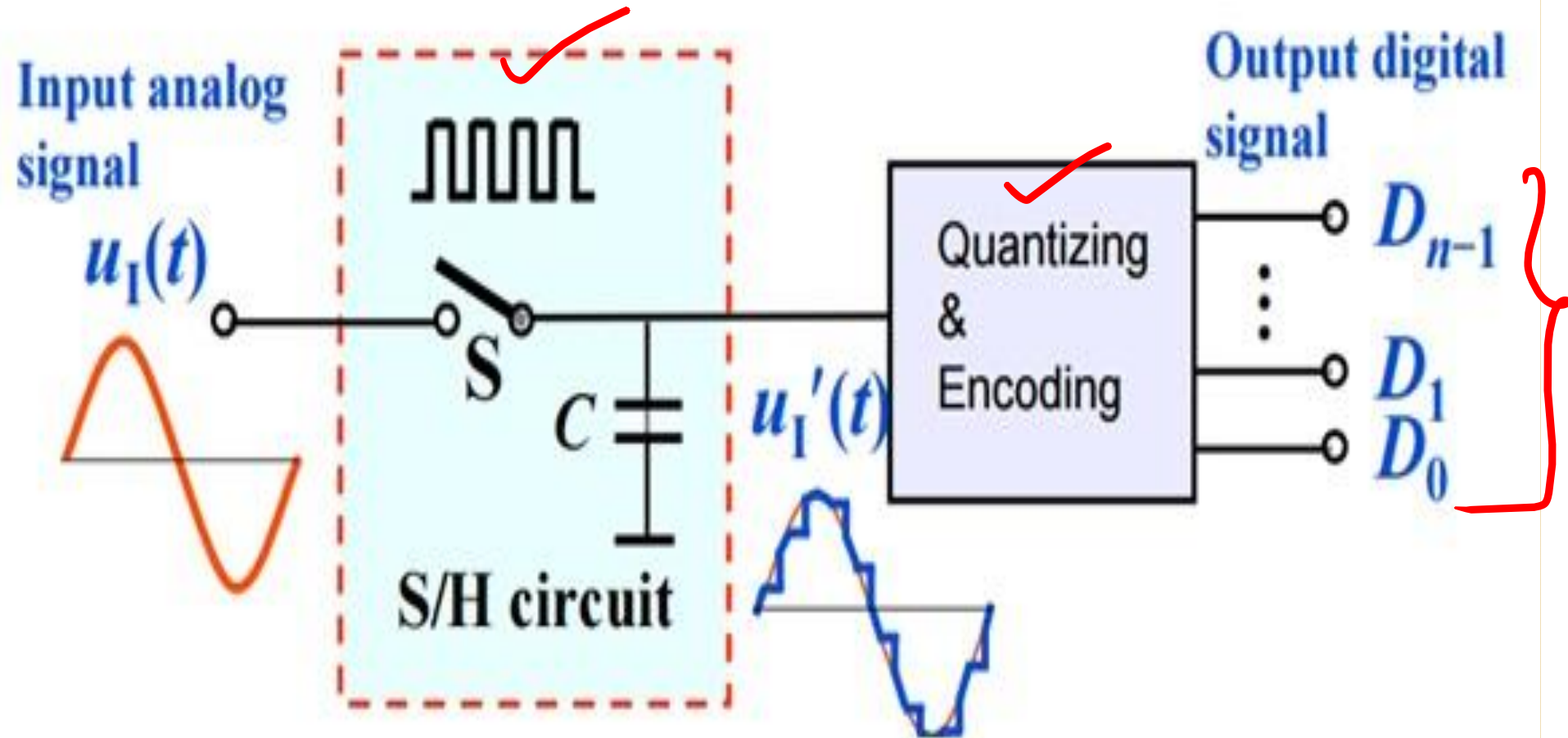
- example: telephone system
 - human voice $\leftrightarrow$ analog data $\leftrightarrow$ analog signal ?!
 - analog signal is sensitive to noise, especially over long distance (cannot be perfectly reconstructed)
 - solution:
 - (1) digitize the analog signal at the sender
 - (2) transmit digital signal
 - (3) convert digital signal back to analog data at the receiver



Digital Transmission of Analog signal

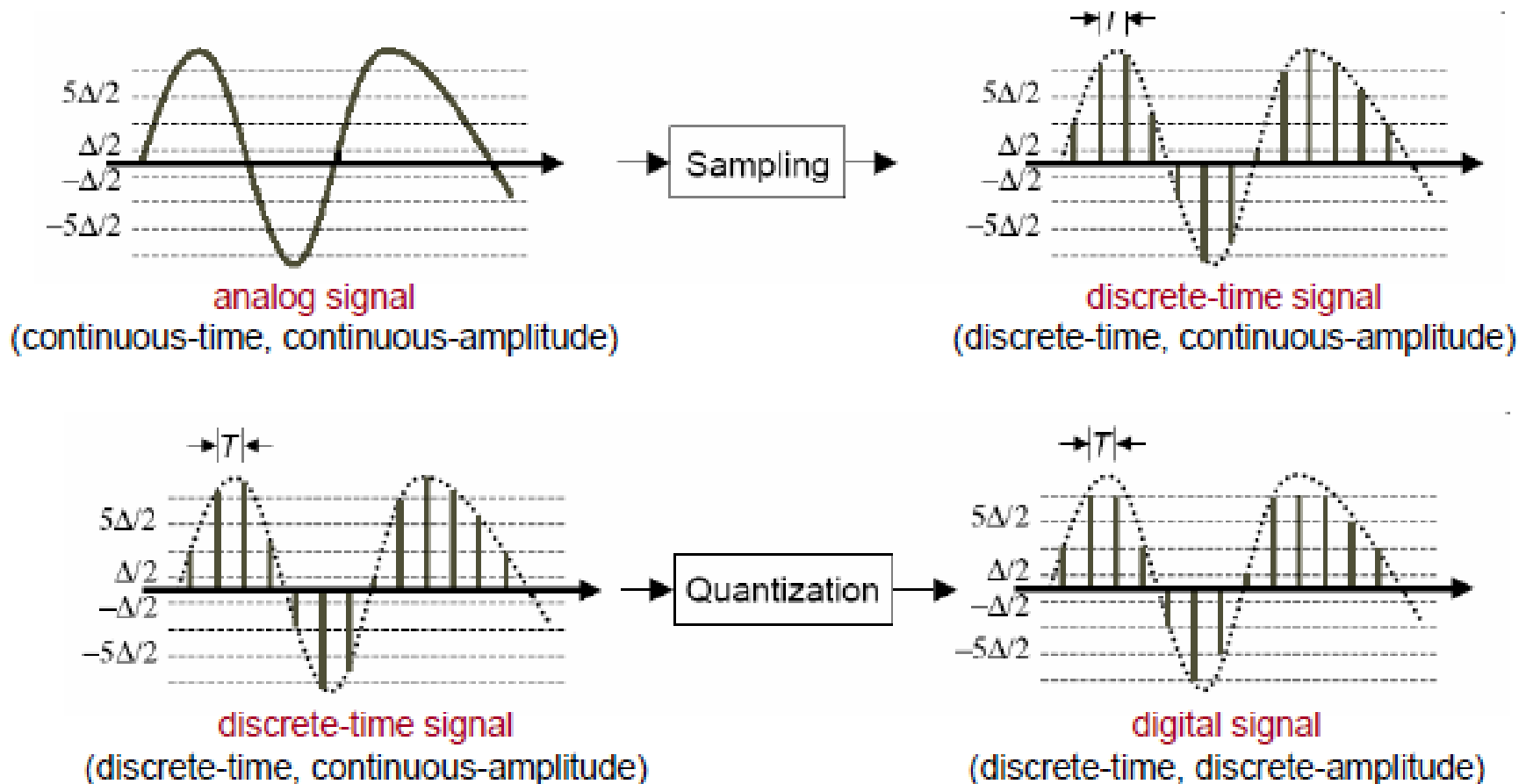


Digital Transmission of Analog signal



Digitization

- Digitization Procedure** – aka **Pulse Code Modulation (PCM)**, consists of 2 steps
- (1) **sampling** – obtain signal values at equal intervals (T)
 - (2) **quantization** – approximate samples to certain values



Pulse Code Modulation (PCM)

- Analog voice data must be translated into a series of binary digits before they can be transmitted.
- With Pulse Code Modulation (PCM), the amplitude of the sound wave is sampled at regular intervals and translated into a binary number.
- The difference between the original analog signal and the translated digital signal is called quantizing error.

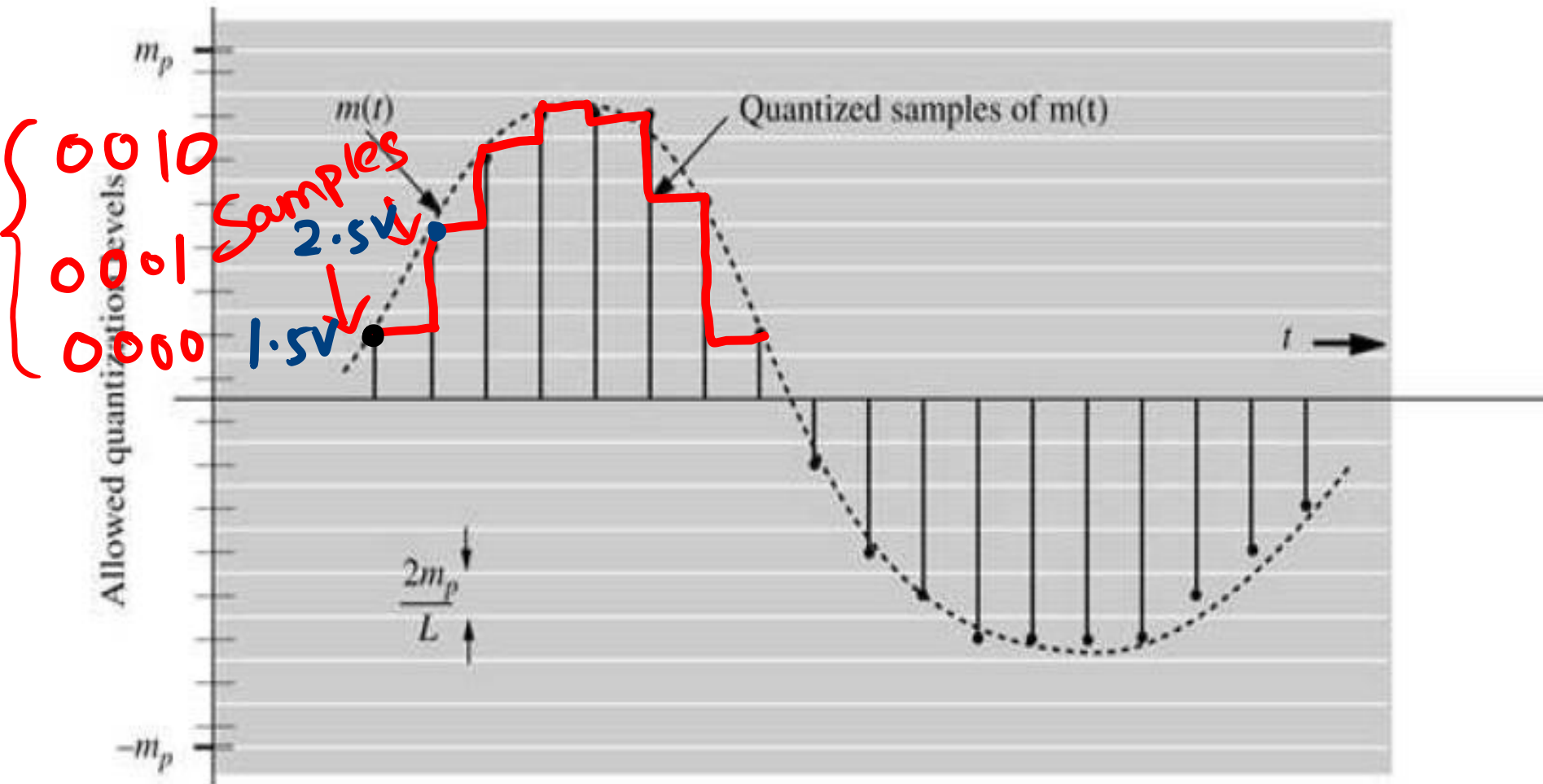
Pulse Code Modulation (PCM)

- PCM uses a sampling rate of 8000 samples per second.
- Each sample is an 8 bit sample resulting in a digital rate of 64,000 bps (8 x 8000).

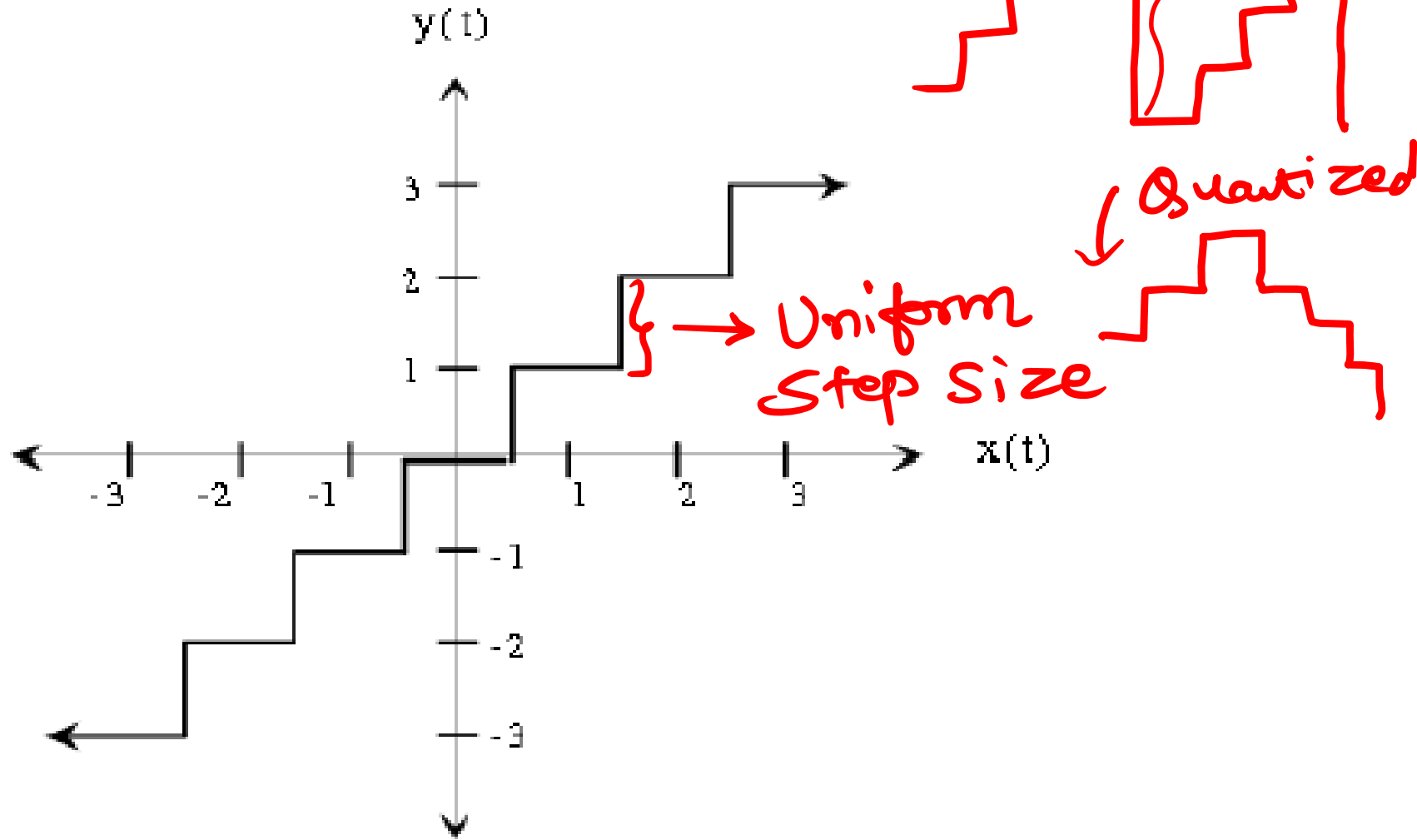
Quantization

Quantization of a signal produces the closest representable value.

For fixed number of values, spacing between values increases with range.



Uniform Quantization



Quantization

- The process of **measuring the numerical values** of the **samples** and giving them a table value in a suitable scale.
- The finite number of amplitude intervals is called the '**quantizing interval**' like quantizing interval no. 1 is 10-20mV; 2 is 20-30mV etc. in a case of 1V signal.
- **Linear quantizing** is where the **quantizing intervals** are of the same size

Quantization

- Quantization intervals are coded in binary form, and so the **quantization intervals** will be in powers of 2.

$$2^8 = 256$$

↓

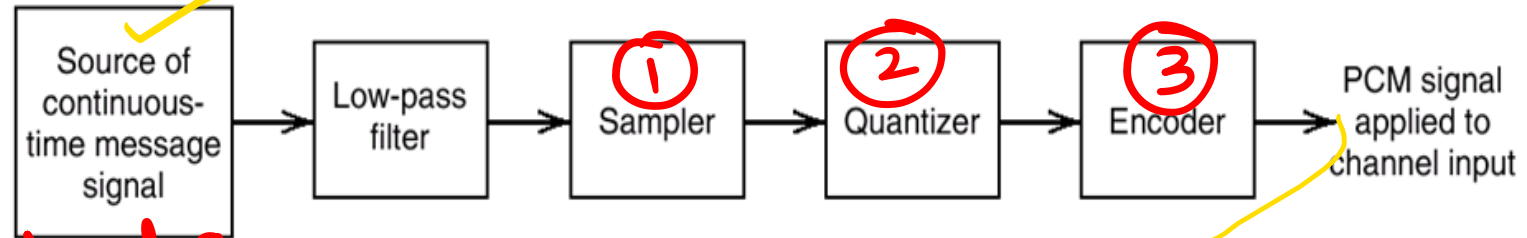
- In PCM, 8 bit code is used and so we have 256 **intervals** for quantizing (128 levels in the positive direction and 128 levels in negative direction)

Quantization Error

- The **deviation between** the amplitude of samples at the transmitter and receiving ends.
- In linear quantization, the **distortion is more** and to decrease the distortion, the no. of steps in the given **amplitude range has to be increased.**
- Due to BW limitations, more quantum levels in small amplitude region are planned results to, **Non linear (uniform) quantization**

PCM Transmission System — Diag. ①

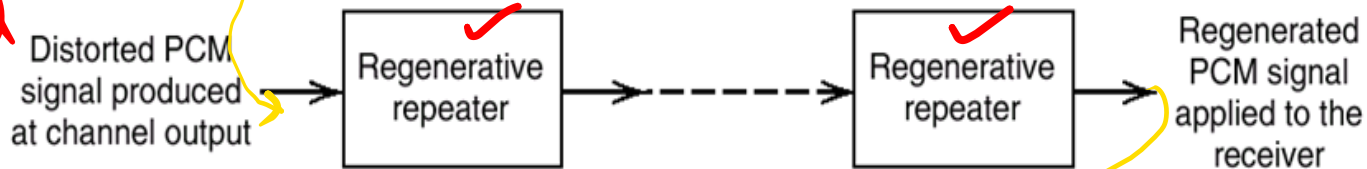
Tx



Analog

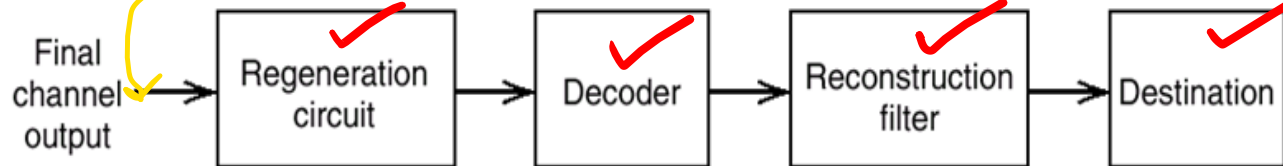
(a) Transmitter

Channel



(b) Transmission path

Rx



(c) Receiver

PCM Transmission System - Diag. ②

